LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 9: Project: Guess the Animal

Module 3 | 50 minutes | Project: lists game | Grades 3–5 · ages 9–11

Objective: Plan, build and test a guessing game that picks a secret from a list and lets the player guess until they get it.

Success criteria

- My game picks a random secret from a list.
- It loops until the guess is right and counts guesses.
- It was playtested.

Key vocabulary

- **secret**: A value the game hides from the player.
- **condition-controlled loop**: A loop that repeats until something is true.
- **counter**: A variable that counts (like guesses).

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Guess my card (5 min): a learner secretly picks a card from four; the class guesses until right, counting guesses.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Plan the game. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Guess the Animal.

Unplugged warm-up

Guess my card (5 min): a learner secretly picks a card from four; the class guesses until right, counting guesses.

Teaching the idea

Combines lists (random selection, membership), variables to hold a fixed secret, and a condition-controlled loop with user input. Note that the secret must be stored before the loop.

Analogy: Twenty Questions: the secret is chosen once, then guesses keep coming.

Live demo: Show the bug where secret is picked inside the loop (it changes every guess!). Move it out.

Questions to ask

Ask	Listen for
Why store the secret in a variable?	So it stays the same during guessing.
When does the loop stop?	When answer = secret.
How could you give clues?	Check contains, or give a hint list.

Differentiation

Support
Provide the list and secret; learners build the loop and message.

Stretch
A hints list where item n matches animal n; say the hint after two wrong guesses.

Common misconceptions

Misconception	How to fix it
pick random inside the loop keeps the same secret	It picks a new one each time; store it first.
repeat until checks only once	It checks before every repeat.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Lists	No list	List but not used	Random secret from the list	List also used for clues or checks
Loop & logic	No loop	Asks once	Repeats until right	Handles odd answers kindly
Variables	None	Secret only	Secret and guesses	Extra like best score

Player experience	Confusing	Works	Clear and friendly	Clues, sounds and polish
Testing & reflection	Not tested	Tested alone	Playtested; improvement explained	Several improvements explained

Exit ticket
Where in your code is the secret chosen, and why there?

Homework
Play your game with someone at home. What was their best number of guesses?

Answer key — Kitabash activities

1. Plan the game (perform · ordering)

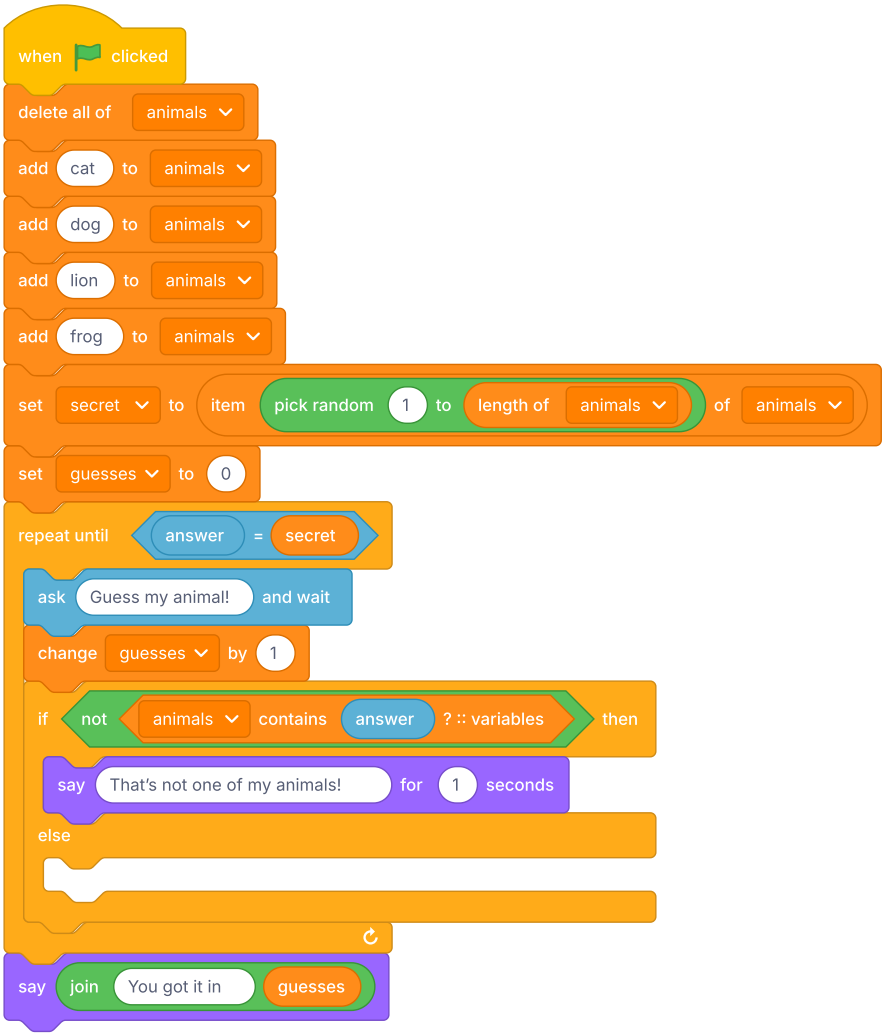
Order: Fill the animals list → Pick a random secret animal → Set guesses to 0 → Ask for a guess until it equals the secret → Say how many guesses it took

2. Project: Guess the Animal (probe · code lab)

Goals checked: 4+ animals in the list · Random secret from the list · Asks for guesses · Player can win

Example project:

Host



Answer key — worksheet

1. **Key words:** secret = A value the game hides from the player. · condition-controlled loop = A loop that repeats until something is true. · counter = A variable that counts (like guesses).
2. **Plan the game:** 1 Fill the animals list · 2 Pick a random secret animal · 3 Set guesses to 0 · 4 Ask for a guess until it equals the secret · 5 Say how many guesses it took

3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.